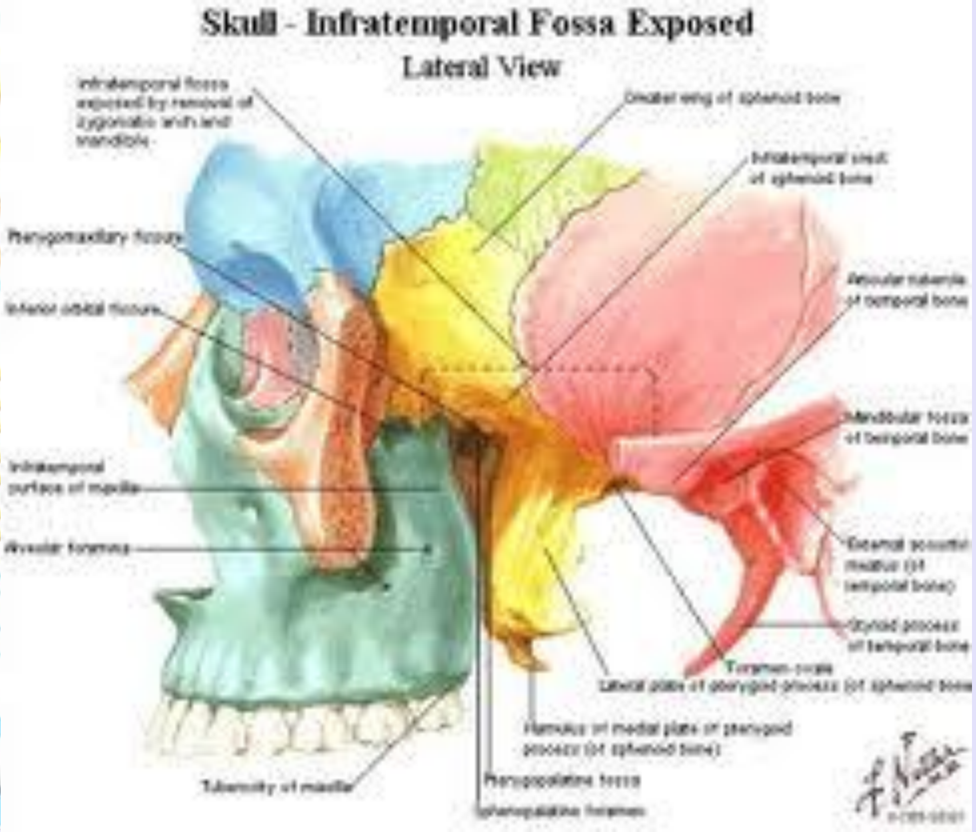
INFRATEMPORAL FOSSA

Learning objectives

- At the end of the lecture the student should be able to know:
- Infratemporal fossa
- Boundaries of infratemporal fossa.
- Contents of infratemporal fossa.
- Maxillary artery and its branches
- Mandibular nerve and its branches.

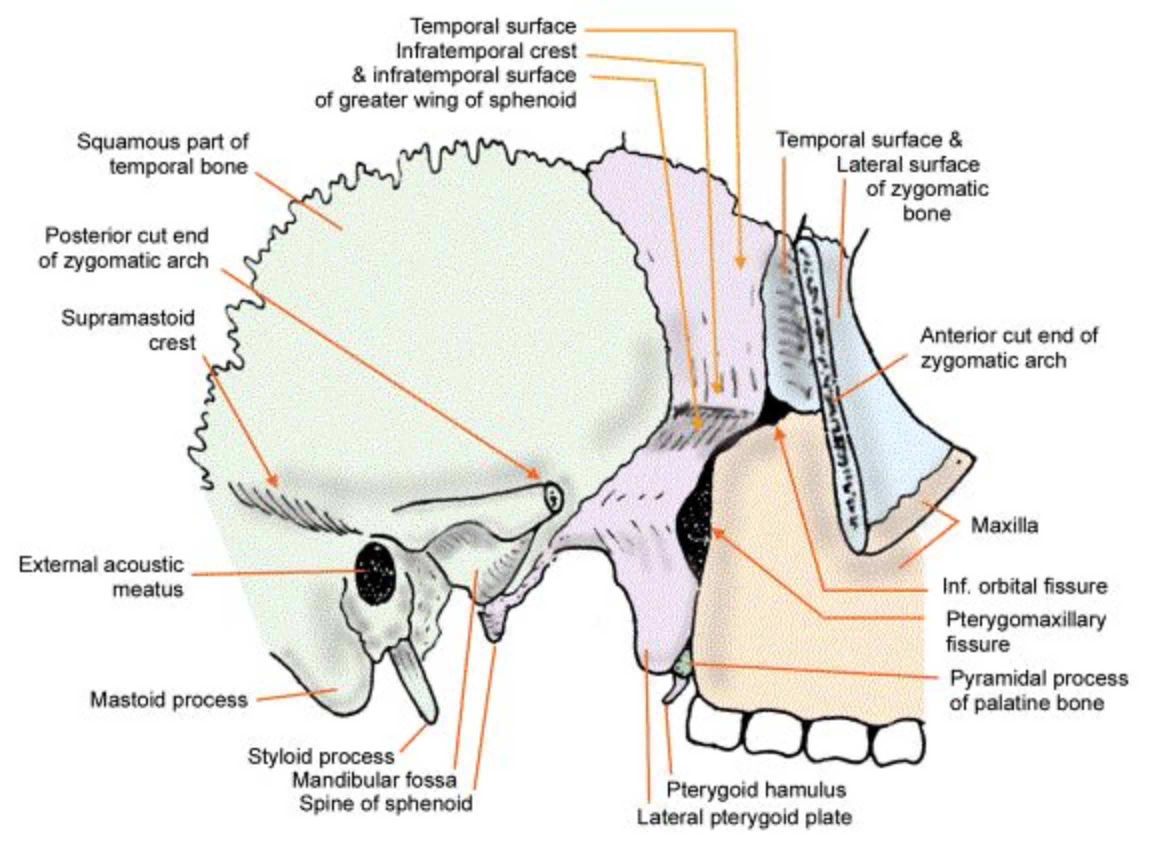
Infratemporal fossa

- The **infratemporal fossa** is an irregularly shaped cavity, situated below and medial to the zygomatic arch.
- The **infratemporal fossa** lies below the **infratemporal crest** on the greater wing of the sphenoid.



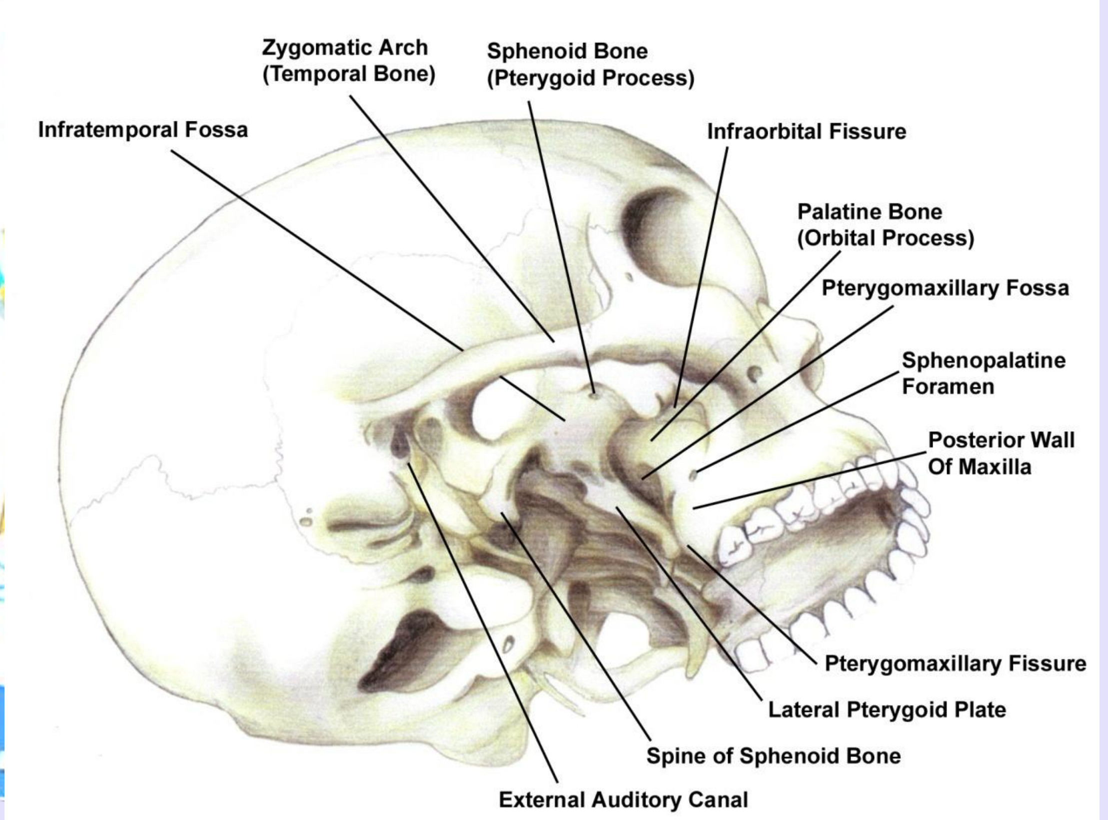
Boundaries

- **Anterior**,
- Infratemporal surface of the maxilla and the ridge which descends from its zygomatic process
- **Posterior**,
- Articular tubercle of the temporal and the spinal angularis of the sphenoid



Boundaries

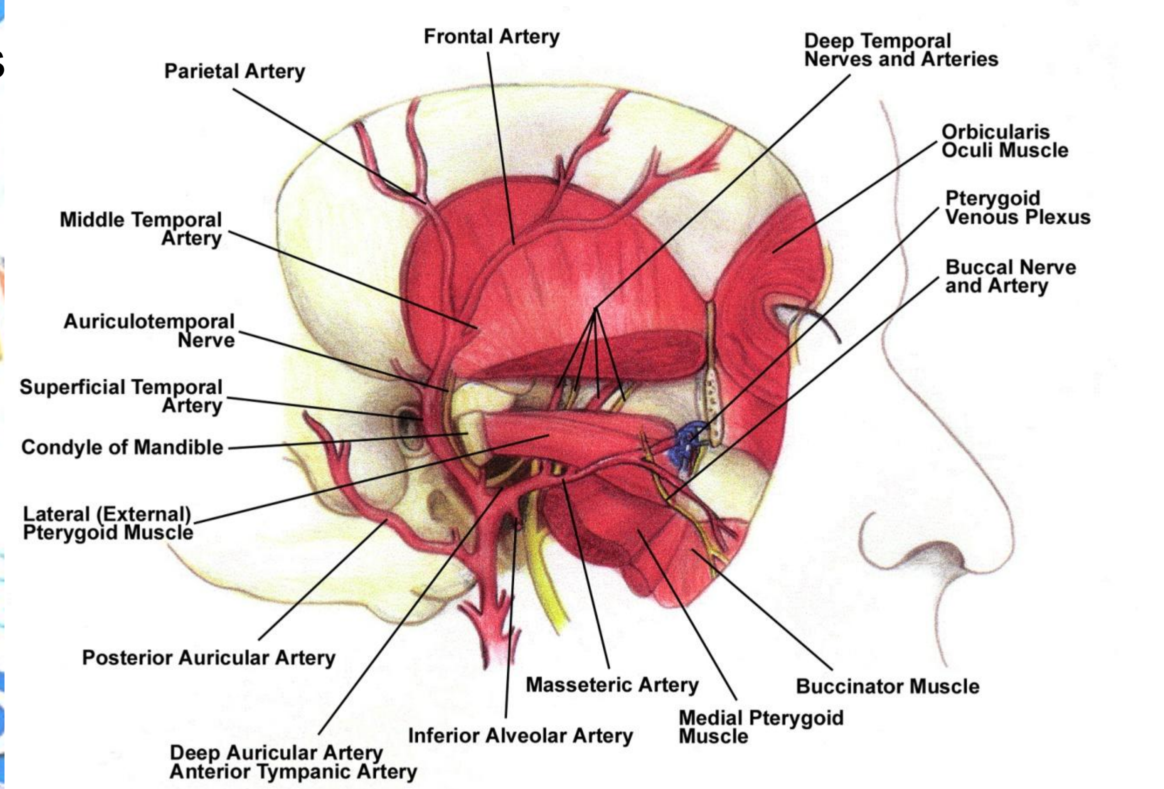
- *Superior*,
  - Greater wing of the sphenoid below the infratemporal crest, and by the under surface of the temporal squama
- Inferior
  - Alveolar border of maxilla.
- Medial
  - Lateral pterygoid plate
- Lateral
  - Ramus of mandible
- Floor is formed by the Medial pterygoid muscle (superior surface where it insets into the mandible)



Osteology of the base of the skull ans the pterygonmaxilly fossa.

Contents of the infratemporal fossa

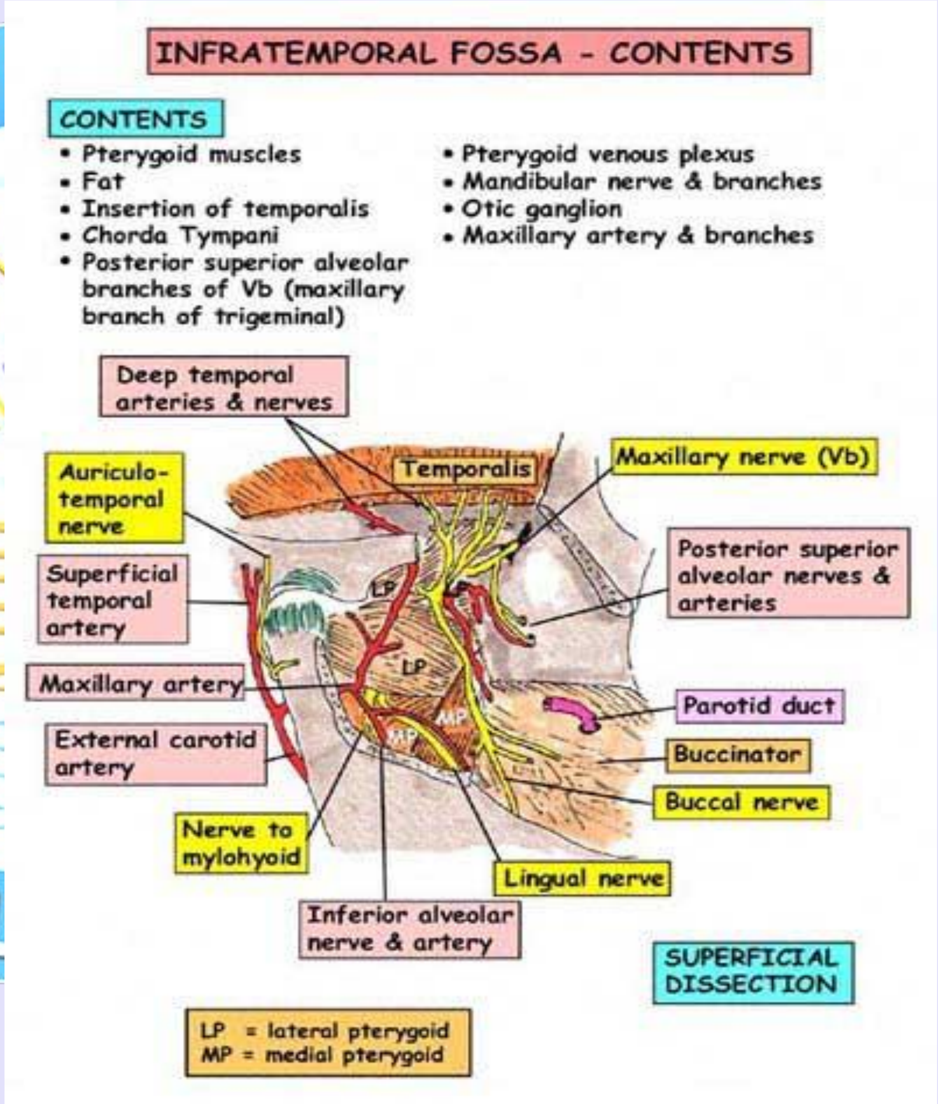
- **Muscles.**
  - Lower part of temporalis
  - Lateral and medial pterygoid
- **Vessels**
  - Maxillary artery originating from the external carotid artery and its branches.
- **Veins**
  - pterygoid venous plexus



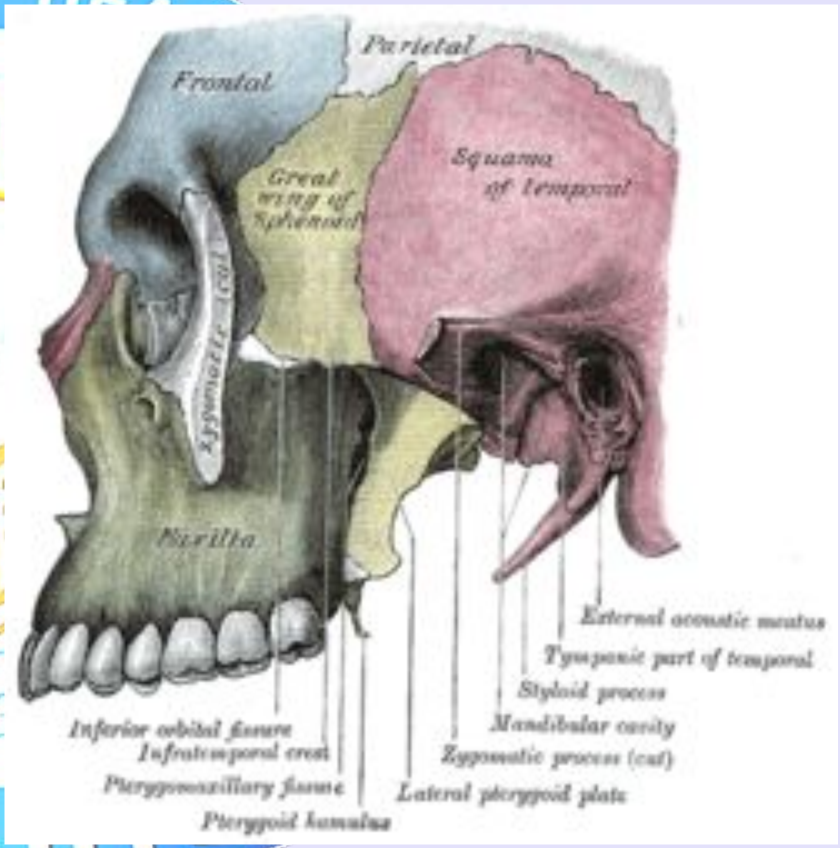
Contents of the infratemporal fossa.

Contents of infratemporal fossa

- **Nerves**
  - Mandibular nerve,
  - Inferior alveolar nerve
  - Lingual nerve
  - Buccal nerve
  - Chorda tympani nerve
  - Otic ganglion

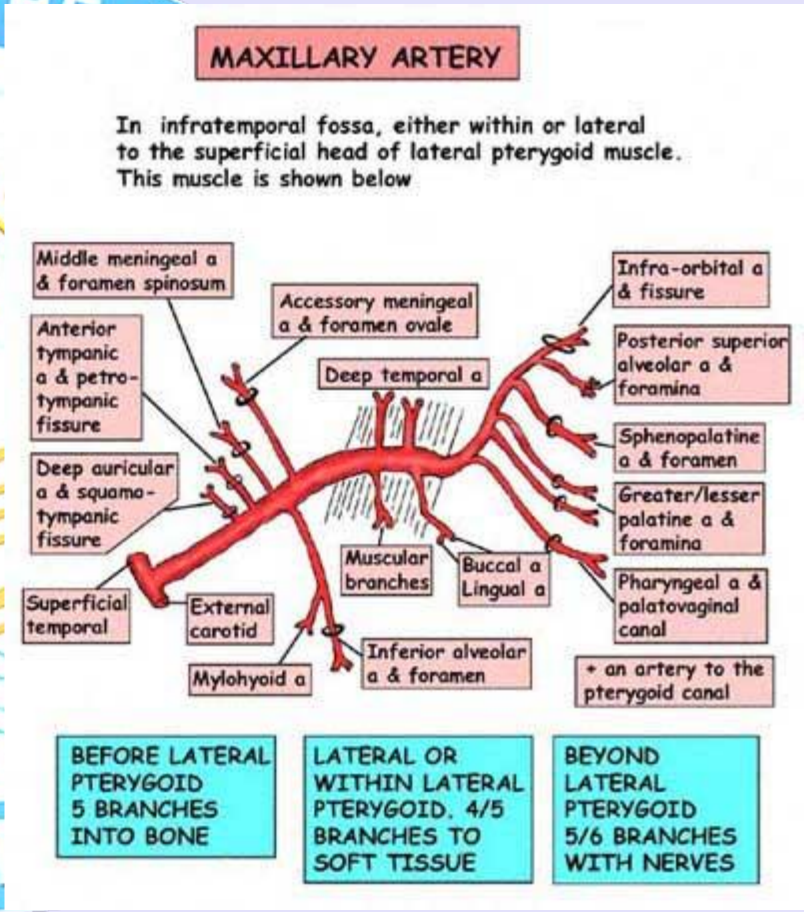


- The **foramen ovale** and **foramen spinosum** open on its roof, and the **alveolar canals** on its anterior wall.
- Two fissures, Inferior orbital, and the vertical one the pterygomaxillary forming a T-shaped fissure at its upper and medial part.



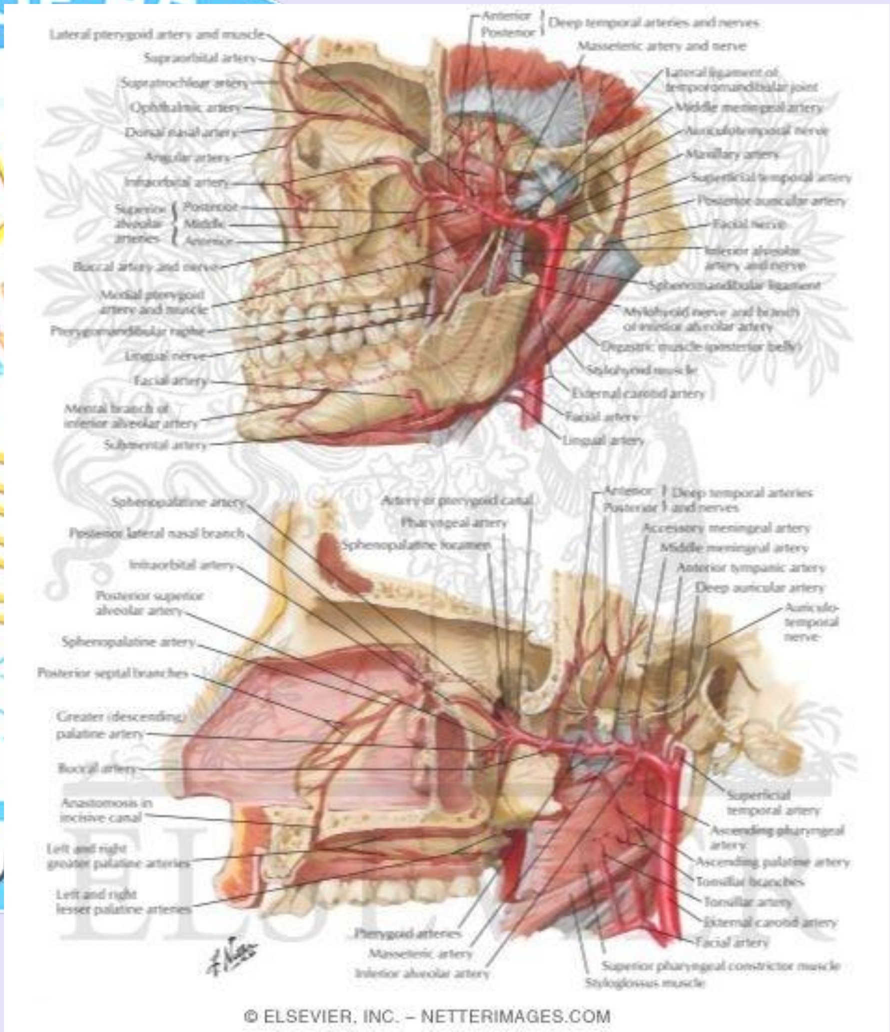
Maxillary artery

- Branch of external carotid
- Arises behind mandible.
- Supplies the deep structures of the face, and may be divided into
- **Mandibular**
- **Pterygoid**
- **Pterygopalatine portions.**



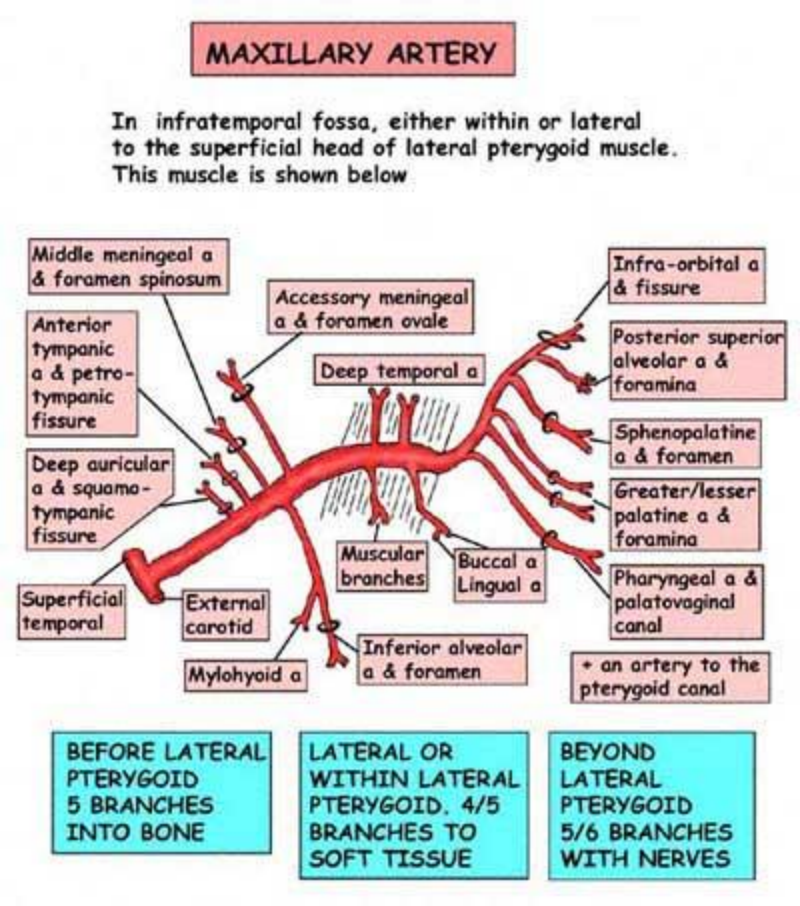
Maxillary artery

- First portion
- passes horizontally forward, between the ramus of the mandible and the sphenomandibular ligament, where it lies parallel to and a little below the auriculotemporal nerve; it crosses the inferior alveolar nerve, and runs along the lower border of the lateral pterygoid.
- **Branches.**
- Middle meningeal artery
- Inferior alveolar artery



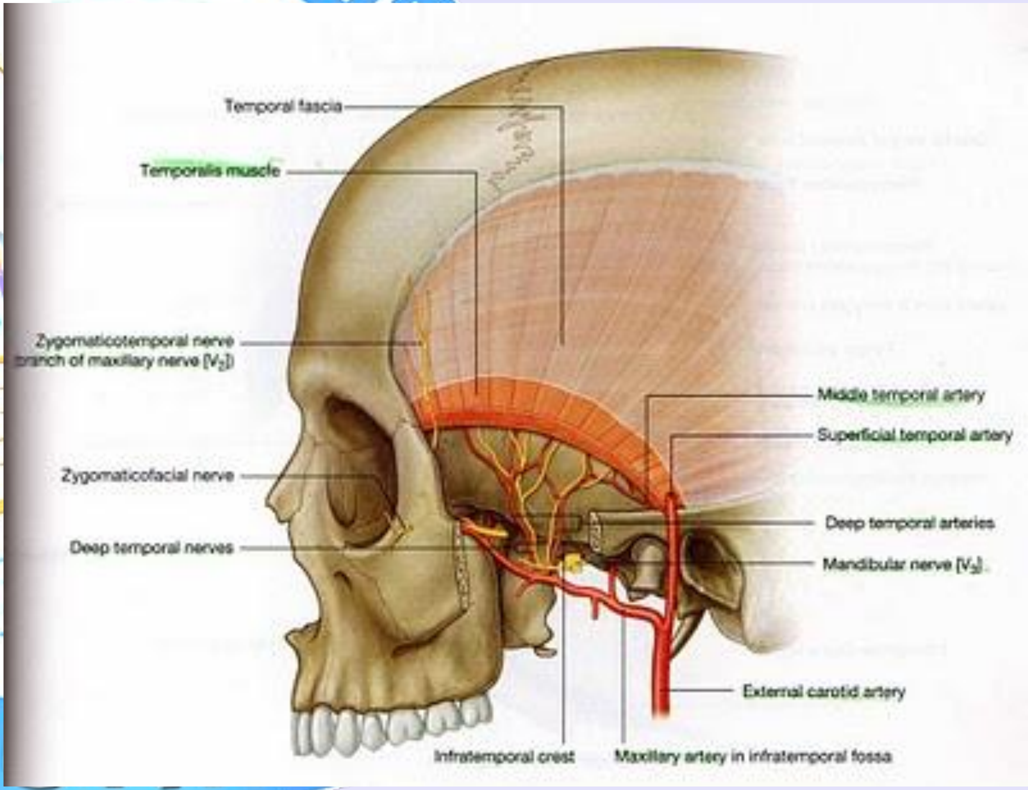
Maxillary artery

- Second portion.
- runs obliquely forward and upward under cover of the ramus of the mandible and insertion of the Temporalis, on the superficial surface of the Pterygoideus externus; it then passes between the two heads of origin of this muscle and enters the fossa.
- **<u>Branches.</u>**
- Deep Temporal.
- Masseteric.
- Pterygoid.
- Buccal.



Maxillary artery

- Third portion
- Lies in the pterygopalatine fossa in relation with the sphenopalatine ganglion.
- **<u>Branches.</u>**
- Posterior Superior Alveolar.
- Artery of the Pterygoid Canal.
- Infraorbital.
- Pharyngeal.
- Descending Palatine.
- Sphenopalatine



Pterygoid venous plexus

- P**terygoid plexus** is a venous plexus of considerable size, and is situated between the temporalis muscle and lateral pterygoid muscle, and partly between the two pterygoid muscles.
- It receives tributaries corresponding with the branches of the maxillary artery
- Thus it receives the following veins:
- sphenopalatine
- middle meningeal
- deep temporal (anterior & posterior)
- pterygoid
- masseteric
- buccinator
- alveolar
- some palatine veins (palatine vein which divides into the greater and lesser palatine v.)
- a branch which communicates with the ophthalmic vein through the inferior orbital fissure
- infraorbital vein

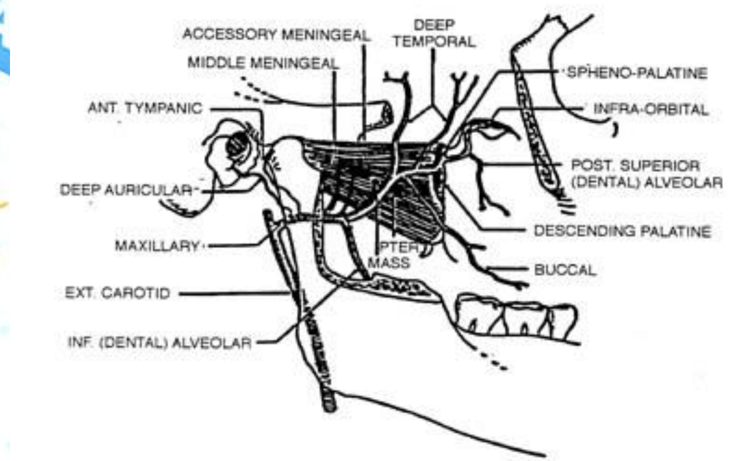


FIGURE 6

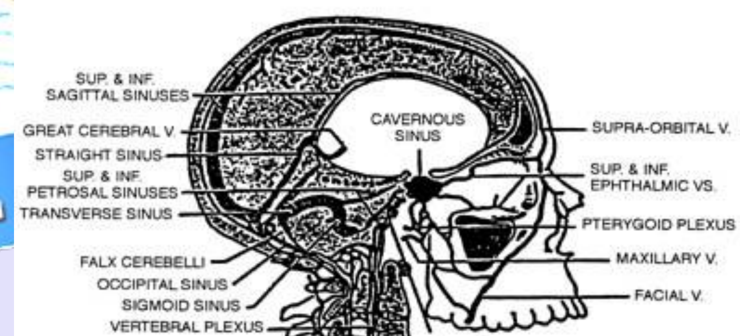
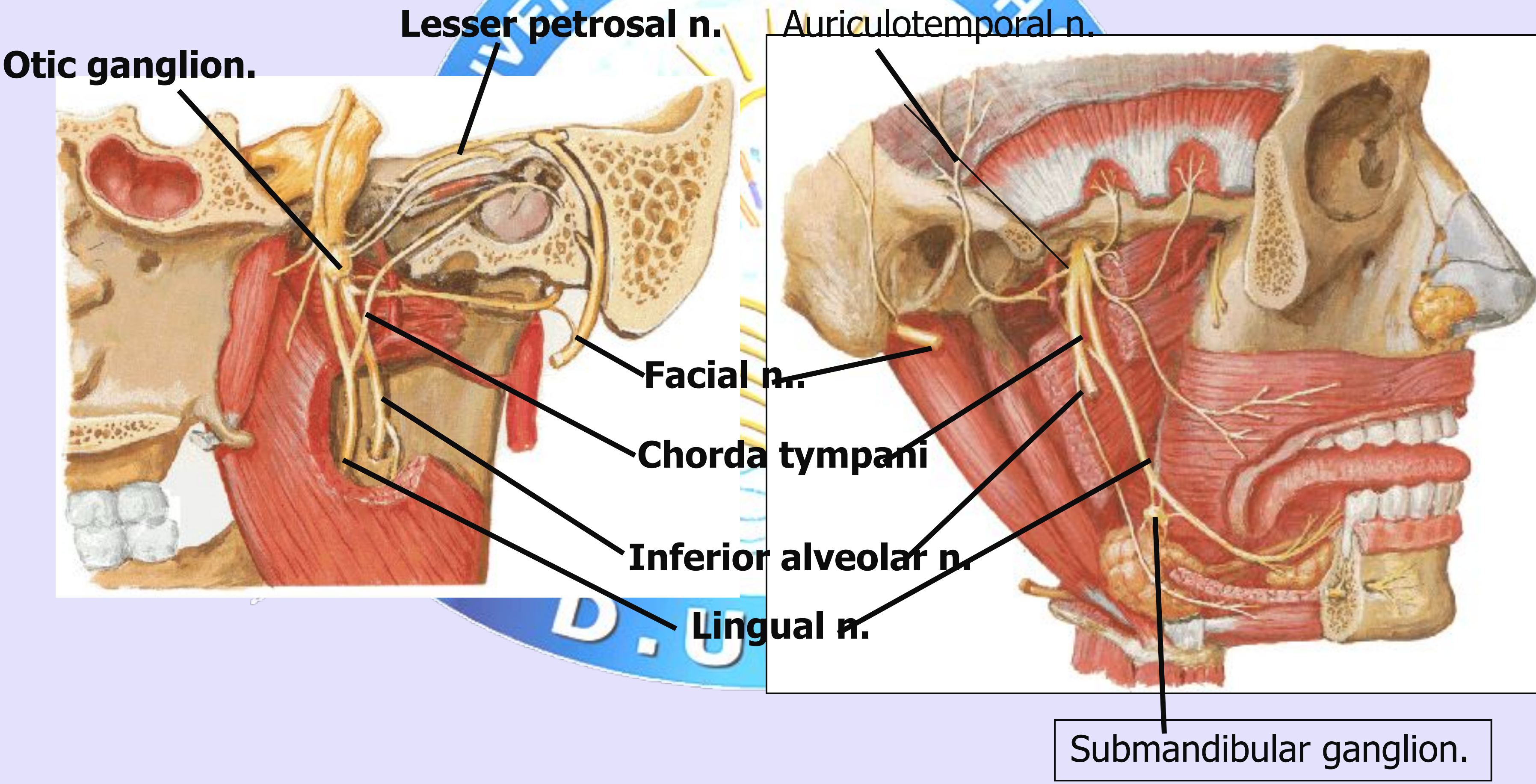


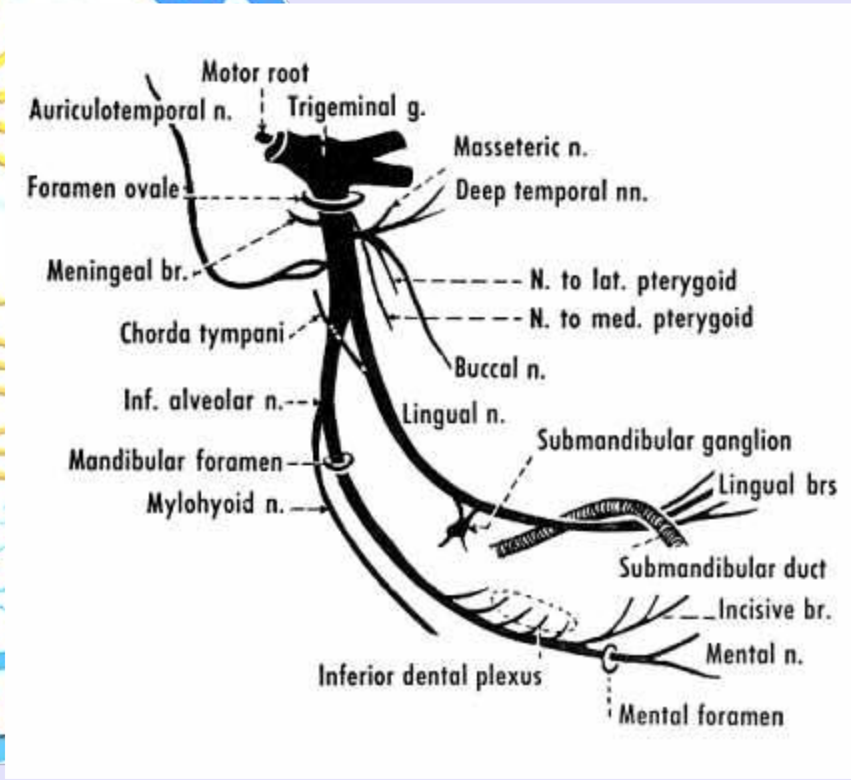
FIGURE 7

Mandibular nerve



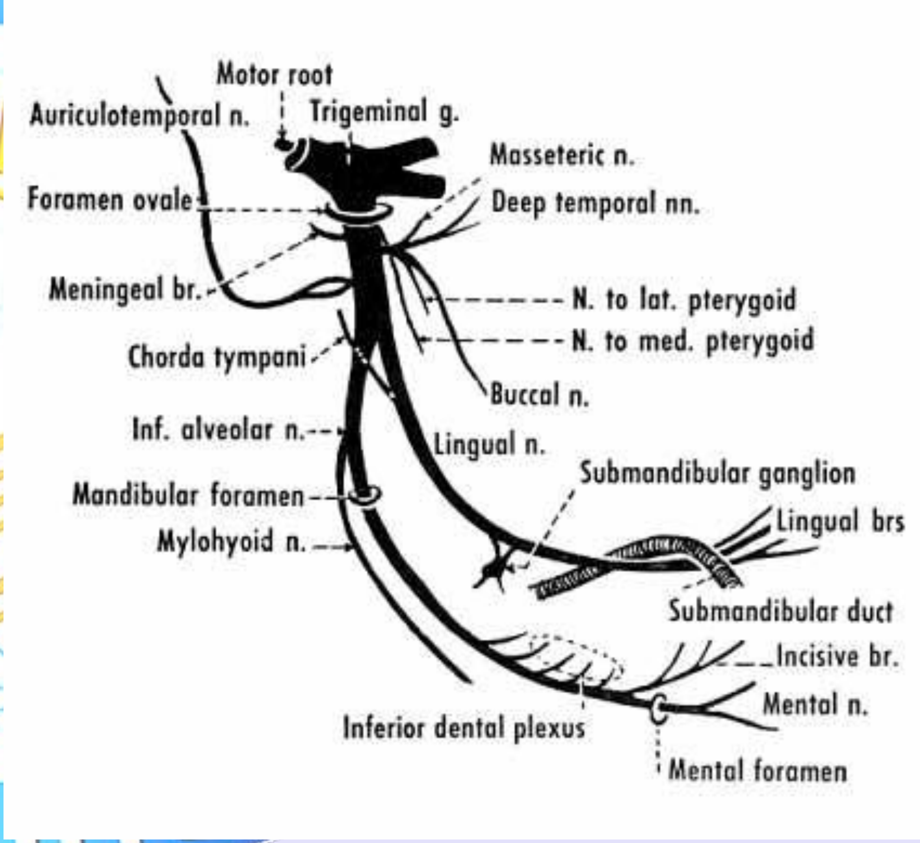
Mandibular nerve

- Largest branch of trigerminal nerve.
- The two roots (sensory and motor) exit the middle cranial fossa through the foramen ovale. The two roots then combine.
- Immediately in the infratemporal fossa beneath the base of the skull, the nerve gives off two branches from its medial side:
- a recurrent branch (nervus spinosus)
- nerve to the medial pterygoid muscle
- Then divides into anterior and posterior division.



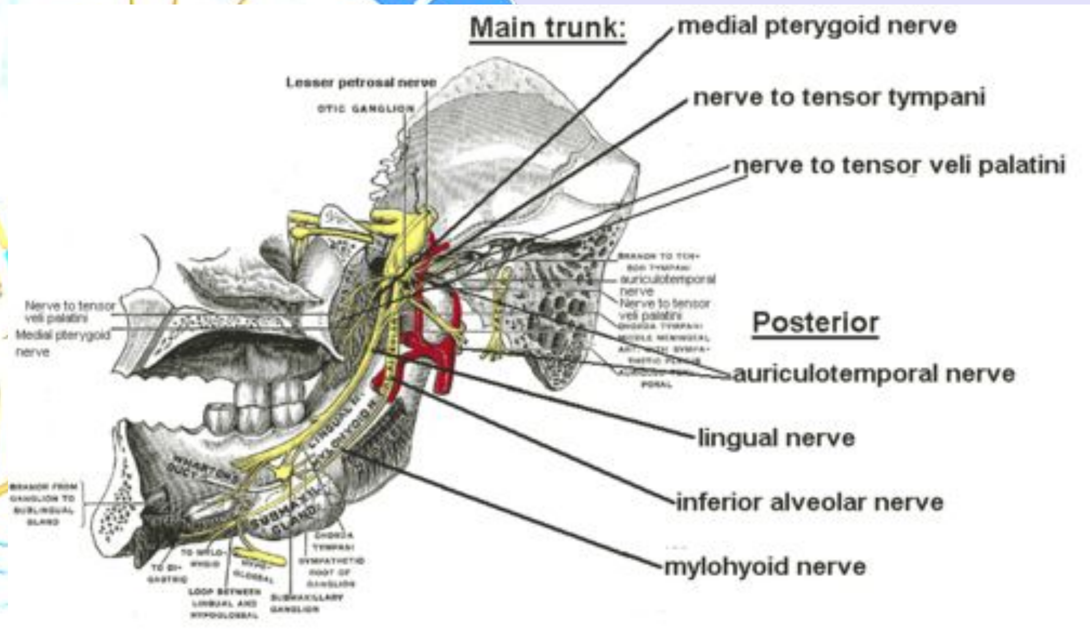
Mandibular nerve

- **From the anterior division**
  - Masseteric nerve
  - Deep temporal nerves (anterior and posterior)
  - Buccal nerve (a sensory nerve)
  - Lateral pterygoid nerve
- Branches from the posterior and anterior divisions (except lateral pterygoid nerve)



Mandibular nerve

- **From the posterior division**
  - Auriculotemporal nerve
  - Lingual nerve
  - Inferior alveolar nerve
  - Motor branch to mylohyoid and anterior belly of digastric muscles (mylohyoid nerve)
- The mandibular nerve also gives off branches to the otic ganglion



Thank you